🌐TWITTER SENTIMENT ANALYSIS USING MACHINE LEARNING

Classifying Twitter Sentiment for Apple and Google Products

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BUSINESS UNDERSTANDING

The Problem

- Millions of customers express opinions online every day.
- Companies need to:
- Monitor brand reputation
- Detect customer dissatisfaction
- Understand public sentiment
- Improve decision-making

Project Goal

- Develop a machine learning model capable of automatically classifying tweets into:
- Positive
- Neutral
- Negative

DATASET OVERVIEW

Dataset Summary

Metric	Value
Source	Twitter
Brands	Apple & Google
Target	Sentiment
Classes	Positive, Neutral, Negative

Features

📝 Tweet Text

🏢 Target Brand

❤️ Sentiment Label

DATA CLEANING & PREPARATION

```
Raw Tweets
          ↓
  Lowercase
          ↓
  Remove URLs
          ↓
  Remove Mentions
          ↓
  Remove Emojis
          ↓
  Remove Stopwords
          ↓
  Lemmatization
          ↓
  Clean Tweets
```

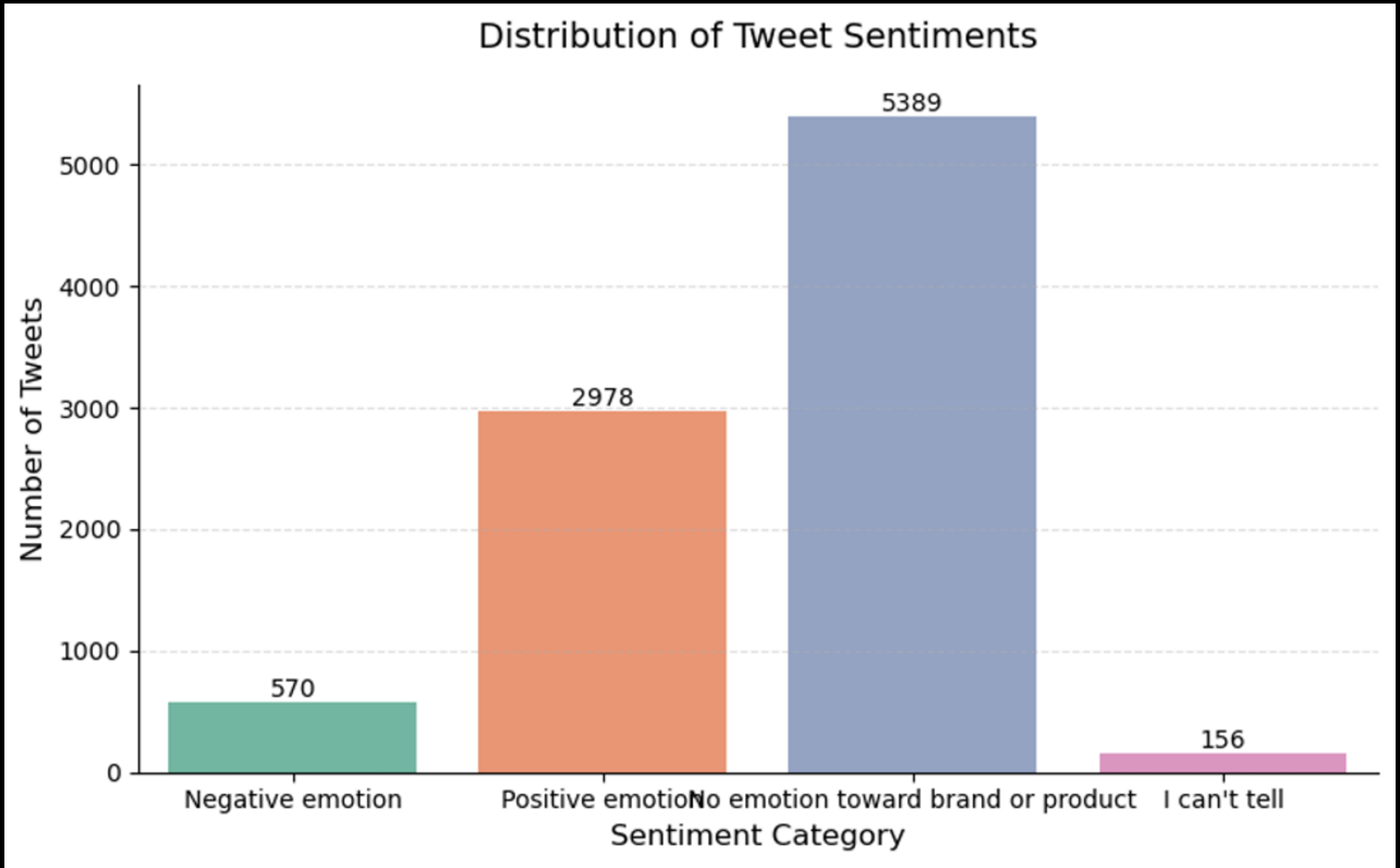
Why?

✅ Reduce noise

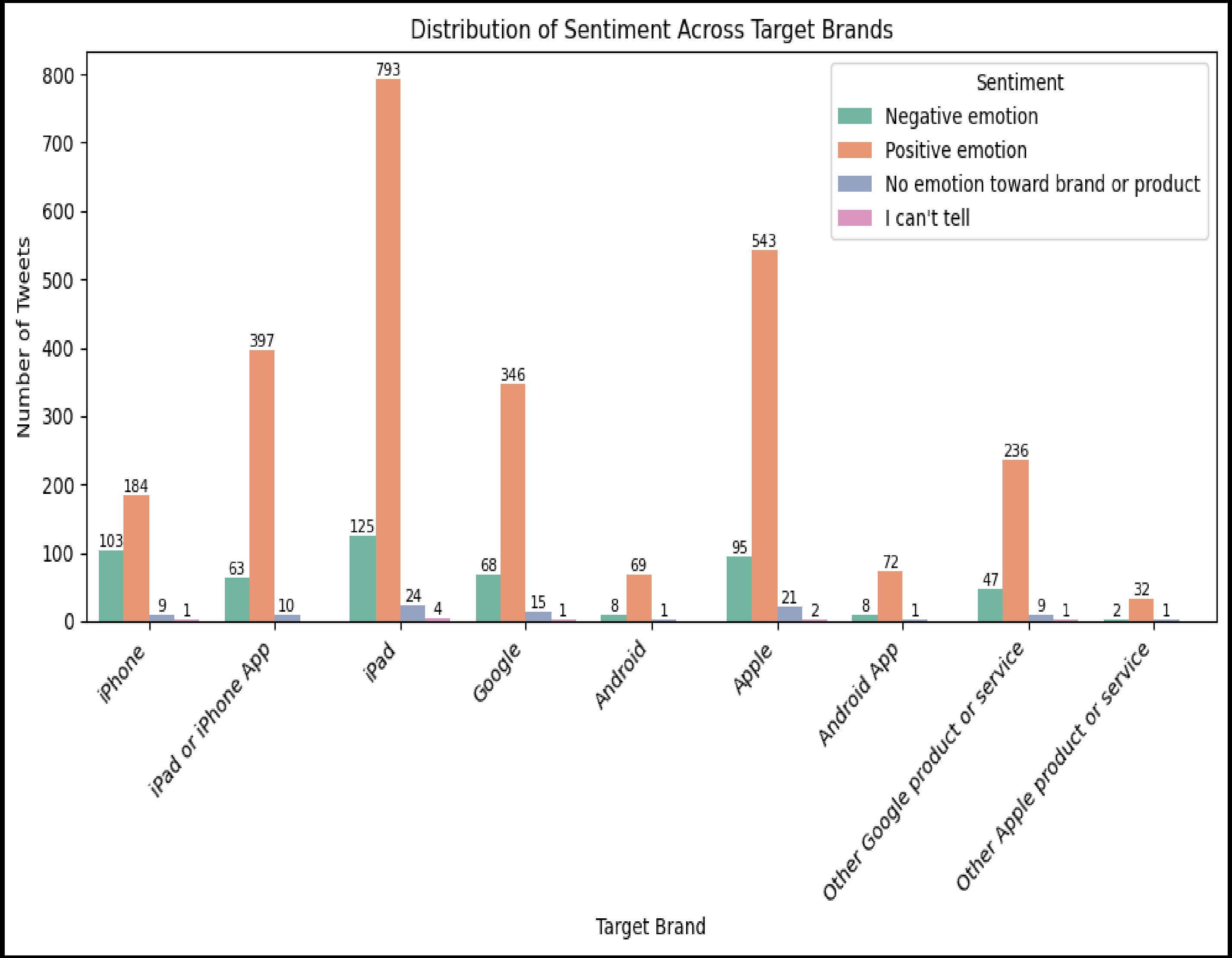
✅ Improve feature quality

✅ Better model performance

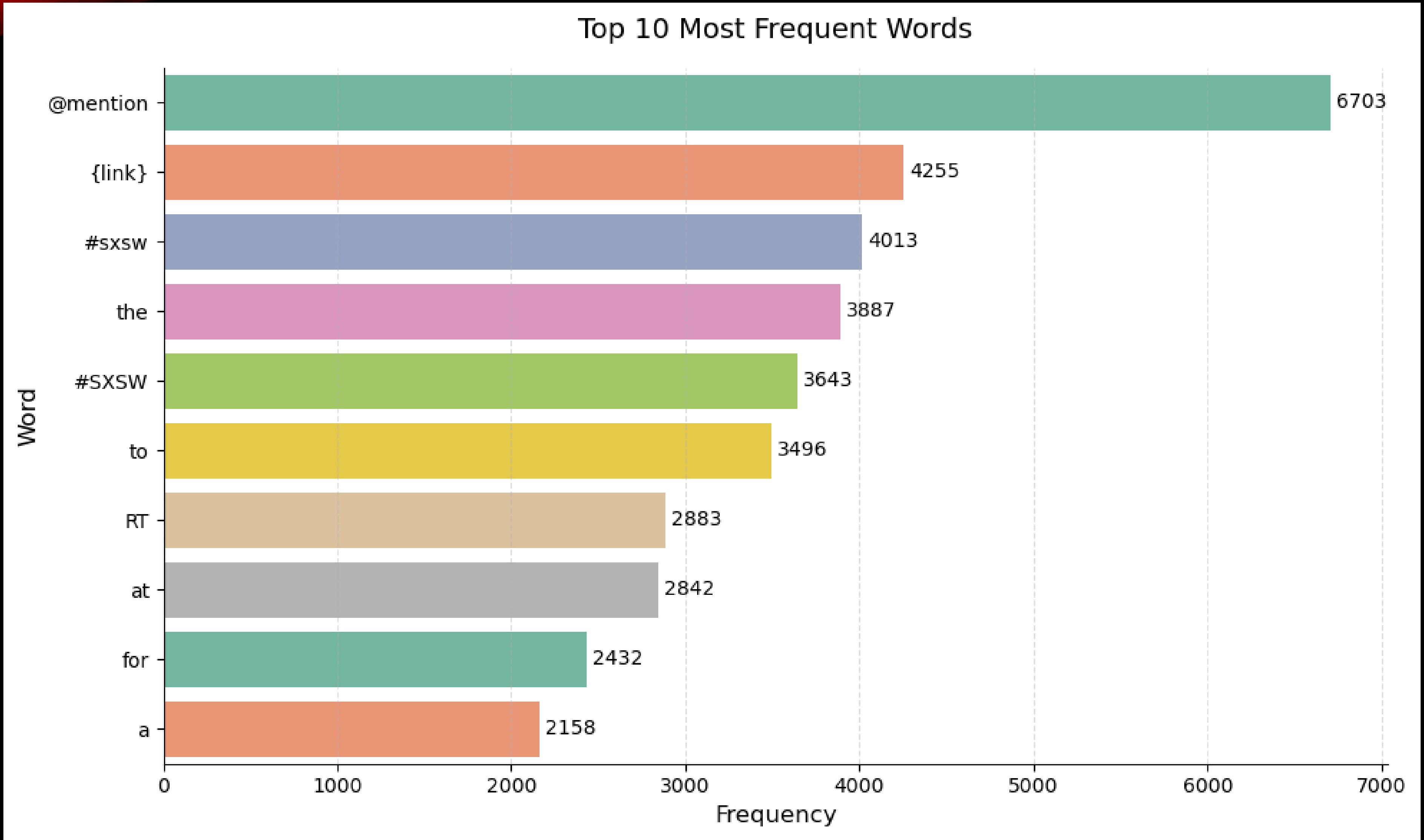
EXPLORATORY DATA ANALYSIS



BRAND ANALYSIS



FREQUENT WORDS



FEATURE ENGINEERING

Transforming Text into Numbers

- **TF-IDF Vectorization**

Converts words into numerical features.

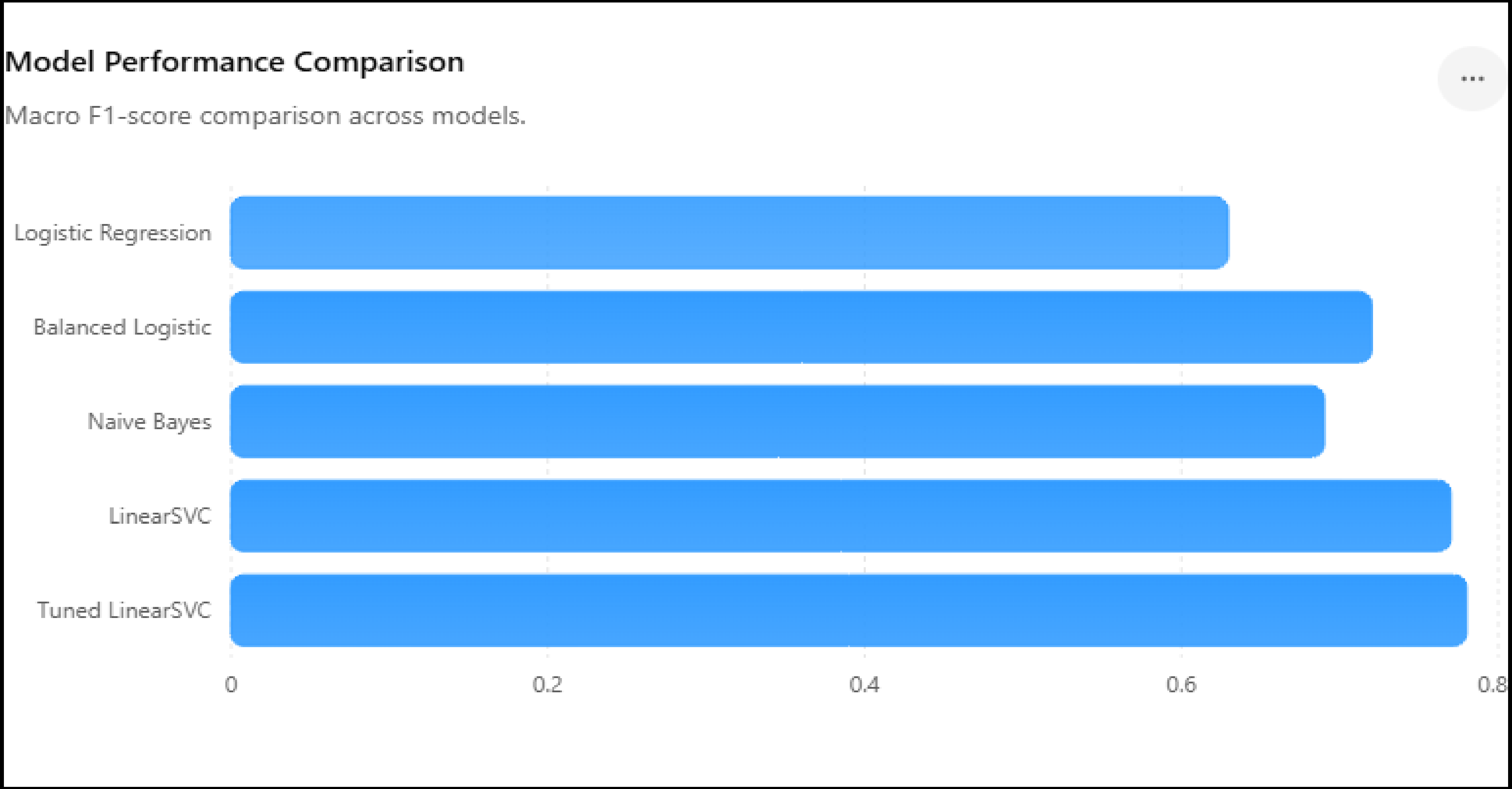
One-Hot Encoding

- Brand Feature:
- Apple → [1,0]
- Google → [0,1]

MACHINE LEARNING MODELS

- **Models Evaluated**
- 🟦 Logistic Regression
- 🟩 Balanced Logistic Regression
- 🟧 Multinomial Naive Bayes
- 🟧 LinearSVC
- 🟥 Tuned LinearSVC
- **Evaluation Metrics**
- Accuracy
- Precision
- Recall
- F1-Score
- Macro F1

MODEL COMPARISON



BEST MODEL RESULTS

- **Tuned LinearSVC**
- Accuracy: **90.6%**
- Macro F1: **0.78**
- **Why It Won**
- ✅ Strong neutral classification
- ✅ Strong positive classification
- ✅ Improved negative sentiment detection

MODEL EXPLAINABILITY

Understanding Predictions

- Feature Importance
- LIME Explanations

Insight

- The model's predictions can be explained using influential words and local explanations, improving transparency and trust.

FINDINGS & RECOMMENDATIONS

Key Findings

- ✅ Most tweets were neutral
- ✅ Negative sentiment was underrepresented
- ✅ Class imbalance affected baseline models
- ✅ Tuned LinearSVC delivered the best performance

Recommendations

- 🚀 Deploy Tuned LinearSVC
- 📈 Collect more negative sentiment examples
- 🔄 Retrain periodically
- 🤖 Explore BERT/Transformer models

THANK YOU

Questions?

Built using Python | Scikit-learn | NLP | Machine Learning